FS Exam Prep — Google Play Launch Roadmap

Version · v1.0.0 — 2026-10-22

App · `FS-Exam-Prep-Mobile-App` (Kotlin / Jetpack Compose), repo HEAD `400b448`

Goal of this roadmap · Get the app onto Google Play **testing tracks** and in front of real users for feedback. Production/monetization is a later phase and is scoped at the end.

Standing decisions applied · Guest-first, no accounts in v1 (2026-10-22) · Calm Pro / Brand Guide v0.2.5 · GUI spec v1.0.0 · publication only under Charlie's own account/brand, with his approval

Readiness verdict (2026-10-22)

We are closer than it feels. No new feature build is required before testing. The app is feature-complete for a feedback build: engine bank connected (480 questions, honest 110-question / 5h 20m mock), 4-tab GUI landed and audited (9/10 §4 checks), de-brand complete, Calm Pro tokens in code, launcher icon implemented (teal adaptive icon). What stands between here and testers is **packaging, accounts plumbing, one real device run, and three small fixes** — not features.

Verified build facts: `applicationId` is currently the AI Studio auto-ID `com.aistudio.fsexamprep.kqzrxp` (source namespace `com.example`); versionCode 1 / versionName "1.0"; minSdk 24, targetSdk 36 (current with Play requirements); the app's own manifest declares no permissions; several unused libraries (Firebase BOM/AI/AppCheck, Retrofit/OkHttp) are bundled.

Phase 0 — Fix & freeze (small Studio pass + 3 decisions)

1. **metadata.json one-liner** — delete the `MAJOR_CAPABILITY_SERVER_SIDE_GEMINI_API` entry (the §4.9 audit FAIL; the only open audit item).

2. **Leaderboard honesty fix** — the Progress leaderboard ranks the user against simulated peers and says nothing about it. Before any external tester sees it: label it (recommended: rename the section to make the sample nature explicit, e.g. "Leaderboard — sample standings" with a line that real rankings arrive with optional sign-in) or hide the row. Labeling is the cheap, honest option; removal is acceptable. *Charlie's call.*
3. **Progress history cap** — decide whether the `take(10)` cap on completed modules stands or reverts to the pre-GUI unbounded list (audit deviation, 2026-10-22). *Charlie's call.*
4. **Final applicationId — decide before any upload.** The current `com.aistudio.fsexamprep.kqzrxp` is publishable but is not his brand, and **an applicationId can never be changed after the first upload**. Pick the permanent one now (it does not need the D1 display name — e.g. a `com.<his-domain-or-handle>.fsprep` form). The `com.example` source namespace can be tidied later; only the applicationId is forever.
5. **Version scheme** — adopt semver from here: the first testing build is `0.9.0` (versionCode increments monotonically).

Gate: repo pushed with 1–3 applied, 4–5 recorded. Nothing else changes — feature freeze for the testing build.

Phase 1 — First real build & device verification (Charlie's PC)

1. Open the project in Android Studio; create a **dedicated upload key** (do not let the AI Studio keystore become the production key by accident) and store it + passwords in his password manager, backed up. Losing the upload key is recoverable via Play support, but it is a bad week.
2. Build a signed **release AAB**; install the APK variant on his phone.
3. **Smoke test (the first device run ever — all audits so far are static):** fresh install → app follows system theme (no forced dark) → Today/Practice/Learn/Progress pass in light AND dark → domain quiz end-to-end → mock exam start, palette, flagging → flashcards → quit and relaunch: streak, daily goal, SM-2 review state all persist → disclaimer visible on Progress.
4. Fix what the smoke test finds (Studio passes, re-audit on push).

Gate: one crash-free full pass on a real device.

Phase 2 — Play Console & listing (Charlie's accounts; Muse drafts)

1. **Google Play Console developer account** — $25 one-time; identity verification can take days, so start early. Personal account (simplest; an organization account needs a D-U-N-S number and exposes a business address).
2. Create the app entry with the Phase 0 applicationId; enroll in **Play App Signing** (Google holds the app key; his key is the upload key).
3. **Store listing v1** (Brand Guide §4 template): 512×512 icon (the decided C3 FS mark), 1024×500 feature graphic, ≥ 2 phone screenshots (aim 4–6: Today, quiz, results, Learn), short + full description in the guide's claim order — diagnostic → bank depth → spaced repetition → worked solutions → offline — with the non-affiliation disclaimer in the full description.
4. **Privacy policy** — required. Guest-first/no-accounts makes it short and true: no accounts, no data collection, everything on-device. Host on GitHub Pages; Muse drafts the text.
5. **Declarations:** Data safety = no data collected/shared (true once the Phase 5 dependency diet lands; declare conservatively until then), content rating questionnaire, target audience = adults / not designed for children, ads = none, app access = all features available without credentials (guest-first pays off here — no review credentials needed).

Gate: console setup complete; listing saved as draft.

Phase 3 — Internal testing track (first upload)

1. Upload the AAB to the **internal track** (up to 100 testers, near-instant availability, no review gate for the first look).
2. Read the **pre-launch report** (automated Firebase Test Lab crawl: crashes, performance, accessibility, screenshots across devices) and **Android vitals**.
3. Charlie + 1–3 trusted testers use it for a few days.

Gate: pre-launch report clean of crashes; vitals sane.

Phase 4 — Closed test: the feedback phase (the point of all this)

1. **Recruit ≥ 12 testers.** Google requires new personal accounts to run a closed test with **at least 12 opted-in testers for 14 consecutive days** before production access can even be requested — so this phase is both the feedback engine and a hard production prerequisite. Start the 14-day clock only when the build is stable. Natural pool: UMaine SVT classmates, surveying coworkers, surveying communities — the exact target users.
2. **Feedback plumbing (no accounts needed):** a Google Form linked from the Progress tab ("Send feedback"), Play reviews, and 4 predefined questions the feedback must answer: (a) Did the diagnostic feel accurate? (b) Question difficulty vs the real FS? (c) Any wrong answers/explanations spotted? (d) What almost made you stop using it?
3. Ship 1–2 fix builds during the window; keep a public changelog per his versioning convention.

Gate: 14 days × 12 testers complete; feedback triaged into fix / tune / post-launch buckets.

Phase 5 — Production readiness (post-feedback; not before testing)

- **Play Billing** for the decided pass model (FS $49 etc.), entitlements built pass-shaped; fail-extension mechanics.
- **Dependency diet** — remove unused Firebase/Retrofit/OkHttp (smaller binary, no merged-permission questions, Data safety becomes trivially "collects nothing").
- **Bank growth** — 480 questions is a fine testing bank; Azimuth's 6,496 is the marketing gap to close over time via the engine's generators, prioritized by Phase 4 feedback.
- **Production icon set** — draw the family as matched vectors (the AI renders are concepts); FS testing does not wait on this.
- Optional Google sign-in phase (per the 2026-10-22 decision) — only if Phase 4 users actually ask for sync/backup.
- Apply for production access; staged rollout (10% → 50% → 100%).

Explicitly NOT required before testing

New features · user accounts · billing · a bigger bank · the D1 brand name (the listing display name can launch as "FS Exam Prep" descriptively and rebrand later — only the applicationId is permanent) · the family vector icon set · the PS/FE/PE apps.